Arnob Talukder

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Experience

Senior Software Engineer Oct 2021 – Present
Lucova Inc. (Nown POS) TypeScript, Java, React, AWS, AngularJS
Promoted from Junior to Intermediate to Senior Software Engineer

AI Analytics

- Architected the AI analytics chat backend from scratch in TypeScript with Hono, Drizzle ORM, PostgreSQL, and Redis
- Reduced unanswered questions from 30% to 0.5% by rebuilding text-to-SQL as an AWS Bedrock tool-use agent
- Cut LLM token usage by 45% per report query by compressing the MySQL schema 85–90% with an AWS Lambda
- Developed labor and net sales reporting in the React 19 Nown Dash manager app, leveraging TanStack Query and Recharts

POS Platform

- Migrated the Java backend from Java 8 to Java 25 across 200+ files, cutting billed CPU usage by 15%
- Eliminated POS menu load delays with two-tier Redis and client-side caching using a stale-while-revalidate strategy
- Rebuilt the Windows POS installer with electron-builder and NSIS, nearly eliminating install support tickets
- Implemented event processing, scheduled meal regulation, and a checkout refactor in the FIIT meal plan integration

Build and Release

- Unified the Java backend, AngularJS client, and Electron app in a pnpm monorepo on Node 24, cutting CI time by 20%
- Automated test, beta, stable, and Electron release pipelines with GitHub Actions, saving 2 engineer hours per release

Leadership

- Championed the Java 25 and monorepo modernization, convincing skeptical leadership with cost and developer time data
- Mentored 2 junior engineers and reviewed 170+ pull requests as the top contributor to the POS codebase since 2024

Freelance Software Developer Feb 2020 – Present
Self-Employed React, React Native, TypeScript, Node.js, PostgreSQL

- Built the Chittagong Association of Canada event platform, tracking $8,400+ in yearly ticket sales with PostgreSQL
- Developed a React Native QR check-in app and Express admin dashboard for event payments, attendance, and reports
- Raised Kennedy Tax & Accounting site visits by 65% and made first contentful paint 2.5× faster with a React website rebuild
- Enabled first-ever online donations for Bangladesh Centre & Community Services via CanadaHelps and PayPal APIs

Projects

Star Ocean 3 Item Creation Calculator talukder0429.github.io/StarOcean3Tools
Crafting cost planner for a PlayStation 2 RPG React, TypeScript, Tailwind CSS, Vite

- Created a TypeScript planner that predicts every item cost from team modifiers across 401 items and 7 professions
- Optimized price overlap detection to O(n) by intersecting precomputed per-item cost sets with the selected costs

Skills

Languages: Java, TypeScript, JavaScript, SQL

Frameworks: React, Node.js, Express, Hono, React Native, Electron, TanStack Query, Drizzle ORM, Tailwind CSS, AngularJS

Tools: AWS (Bedrock, Lambda, S3, SES, CloudWatch), MySQL, PostgreSQL, Redis, GitHub Actions, pnpm

Education

University of Toronto 2020
Honours Bachelor of Science (HBSc), Computer Science Specialist, Software Engineering Stream